
Random processes

In this chapter we begin to study random processes – collections of random variables $(X_t)_{t \in T}$ that are not necessarily independent. In many classical examples of probability theory such as Brownian motion, t stands for time and thus T is a subset of $\mathbb{R}$. But in high-dimensional probability it is important to go beyond this case and allow T to be a general abstract set. An important example is the so-called canonical Gaussian process

$$X_t = \langle g, t \rangle, \quad t \in T,$$

where T is an arbitrary subset of $\mathbb{R}^n$ and g is a standard normal random vector in $\mathbb{R}^n$. We discuss this in Section 7.1.

In Section 7.2, we prove remarkably sharp comparison inequalities for Gaussian processes – Slepian’s, Sudakov-Fernique’s and Gordon’s. Our argument introduces a useful technique of Gaussian interpolation. In Section 7.3, we illustrate the comparison inequalities by proving a sharp bound $\mathbb{E} \|A\| \leq \sqrt{m} + \sqrt{n}$ on the operator norm of a $m \times n$ Gaussian random matrix A .

It is important to understand how the probabilistic properties of random processes, and in particular canonical Gaussian process, are related to the geometry of the underlying set T . In Section 7.4, we prove Sudakov’s minoration inequality which gives a lower bound on the magnitude of a canonical Gaussian process

$$w(T) = \mathbb{E} \sup_{t \in T} \langle g, t \rangle$$

in terms of the covering numbers of T ; upper bounds will be studied in Chapter 8. The quantity $w(T)$ is called Gaussian width of the set $T \subset \mathbb{R}^n$. We study this key geometric parameter in detail in Section 7.5 where we relate it with other notions including stable dimension, stable rank, and Gaussian complexity.

In Section 7.7, we give an example that highlights the importance of the Gaussian width in high-dimensional geometric problems. We examine how random projections affect a given set $T \subset \mathbb{R}^n$, and we find that Gaussian width of T plays a key role in determining the sizes of random projections of T .

7.1 Basic concepts and examples

Definition 7.1.1 (Random process). A *random process* is a collection of random variables $(X_t)_{t \in T}$ on the same probability space, which are indexed by elements t of some set T .

In some classical examples, t stands for *time*, in which case T is a subset of $\mathbb{R}$. But we primarily study processes in high-dimensional settings, where T is a subset of $\mathbb{R}^n$ and where the analogy with time will be lost.

Example 7.1.2 (Discrete time). If $T = \{1, \dots, n\}$ then the random process

$$(X_1, \dots, X_n)$$

can be identified with a *random vector* in $\mathbb{R}^n$.

Example 7.1.3 (Random walks). If $T = \mathbb{N}$, a discrete-time random process $(X_n)_{n \in \mathbb{N}}$ is simply a *sequence* of random variables. An important example is a *random walk* defined as

$$X_n := \sum_{i=1}^n Z_i,$$

where the increments Z_i are independent, mean zero random variables. See Figure 7.1 for illustration.

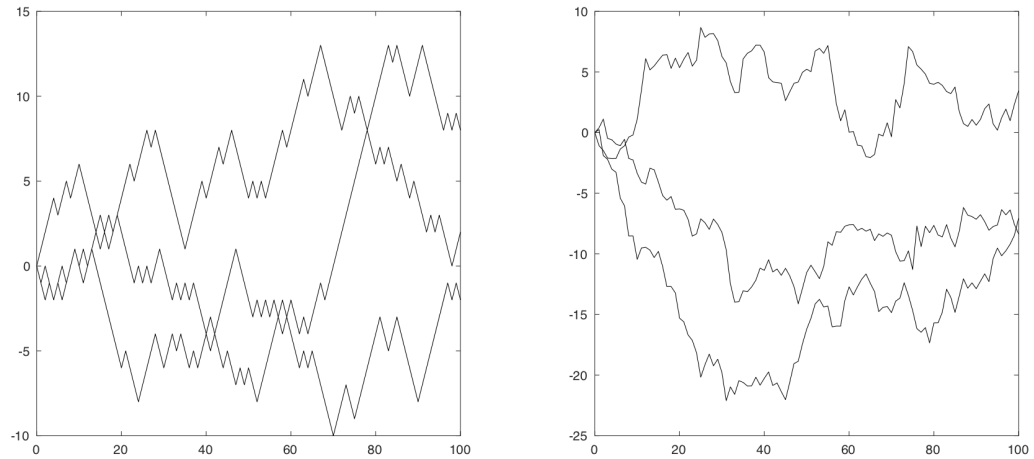


Figure 7.1 A few trials of a random walk with symmetric Bernoulli steps Z_i (left) and a few trials of the standard Brownian motion in $\mathbb{R}$ (right).

Example 7.1.4 (Brownian motion). The most classical continuous-time random process is the standard *Brownian motion* $(X_t)_{t \geq 0}$, also called the *Wiener process*. It can be characterized as follows:

- (i) The process has continuous sample paths, i.e. the random function $f(t) := X_t$ is continuous almost surely;
- (ii) The increments are independent and satisfy $X_t - X_s \sim N(0, t - s)$ for all $t \geq s$.

Figure 7.1 illustrates a few trials of the standard Brownian motion.

Example 7.1.5 (Random fields). When the index set T is a subset of $\mathbb{R}^n$, a random process $(X_t)_{t \in T}$ is sometimes called a spacial random process, or a *random field*. For example, the water temperature X_t at the location on Earth that is parametrized by t can be modeled as a spacial random process.

7.1.1 Covariance and increments

In Section 3.2, we introduced the notion of the covariance matrix of a random vector. We now define the *covariance function* of a random process $(X_t)_{t \in T}$ in a similar manner. For simplicity, let us assume in this section that the random process has zero mean, i.e.

$$\mathbb{E} X_t = 0 \quad \text{for all } t \in T.$$

(The adjustments for the general case will be obvious.) The covariance function of the process is defined as

$$\Sigma(t, s) := \text{cov}(X_t, X_s) = \mathbb{E} X_t X_s, \quad t, s \in T.$$

Similarly, the *increments* of the random process are defined as

$$d(t, s) := \|X_t - X_s\|_{L^2} = (\mathbb{E}(X_t - X_s)^2)^{1/2}, \quad t, s \in T.$$

Example 7.1.6. The increments of the standard Brownian motion satisfy

$$d(t, s) = \sqrt{t - s}, \quad t \geq s$$

by definition. The increments of a random walk of Example 7.1.3 with $\mathbb{E} Z_i^2 = 1$ behave similarly:

$$d(n, m) = \sqrt{n - m}, \quad n \geq m.$$

(Check!)

Remark 7.1.7 (Canonical metric). As we emphasized in the beginning, the index set T of a general random process may be an abstract set without any geometric structure. But even in this case, the increments $d(t, s)$ always define a *metric* on T , thus automatically turning T into a *metric space*.¹ However, Example 7.1.6 shows that this metric may not agree with the standard metric on $\mathbb{R}$, where the distance between t and s is $|t - s|$.

Exercise 7.1.8 (Covariance vs. increments). ☞☞ Consider a random process $(X_t)_{t \in T}$.

- Express the increments $\|X_t - X_s\|_{L^2}$ in terms of the covariance function $\Sigma(t, s)$.
- Assuming that the zero random variable 0 belongs to the process, express the covariance function $\Sigma(t, s)$ in terms of the increments $\|X_t - X_s\|_{L^2}$.

¹ More precisely, $d(t, s)$ is a *pseudometric* on T since the distance between two distinct points can be zero, i.e. $d(t, s) = 0$ does not necessarily imply $t = s$.

Exercise 7.1.9 (Symmetrization for random processes). 🍷🍷🍷 Let $X_1(t), \dots, X_N(t)$ be N independent, mean zero random processes indexed by points $t \in T$. Let $\varepsilon_1, \dots, \varepsilon_N$ be independent symmetric Bernoulli random variables. Prove that

$$\frac{1}{2} \mathbb{E} \sup_{t \in T} \left| \sum_{i=1}^N \varepsilon_i X_i(t) \right| \leq \mathbb{E} \sup_{t \in T} \left| \sum_{i=1}^N X_i(t) \right| \leq 2 \mathbb{E} \sup_{t \in T} \left| \sum_{i=1}^N \varepsilon_i X_i(t) \right|.$$

Hint: Argue like in the proof of Lemma 6.4.2.

7.1.2 Gaussian processes

Definition 7.1.10 (Gaussian process). A random process $(X_t)_{t \in T}$ is called a *Gaussian process* if, for any finite subset $T_0 \subset T$, the random vector $(X_t)_{t \in T_0}$ has normal distribution. Equivalently, $(X_t)_{t \in T}$ is Gaussian if every finite linear combination $\sum_{t \in T_0} a_t X_t$ is a normal random variable. (This equivalence is due to the characterization of normal distribution in Exercise 3.3.4.)

The notion of Gaussian processes generalizes that of Gaussian random vectors in $\mathbb{R}^n$. A classical example of a Gaussian process is the standard Brownian motion.

Remark 7.1.11 (Distribution is determined by covariance, increments). From the formula (3.5) for multivariate normal density we may recall that the distribution of a mean zero Gaussian random vector X in $\mathbb{R}^n$ is completely determined by its covariance matrix. Then, by definition, the distribution of a mean zero Gaussian process $(X_t)_{t \in T}$ is also completely determined² by its covariance function $\Sigma(t, s)$. Equivalently (due to Exercise 7.1.8), the distribution of the process is determined by the increments $d(t, s)$.

We now consider a wide class of examples of Gaussian processes indexed by higher-dimensional sets $T \subset \mathbb{R}^n$. Consider the standard normal random vector $g \sim N(0, I_n)$ and define the random process

$$X_t := \langle g, t \rangle, \quad t \in T. \quad (7.1)$$

Then $(X_t)_{t \in T}$ is clearly a Gaussian process, and we call it a *canonical Gaussian process*. The increments of this process define the Euclidean distance

$$\|X_t - X_s\|_{L^2} = \|t - s\|_2, \quad t, s \in T.$$

(Check!)

Actually, one can realize any Gaussian process as the canonical process (7.1). This follows from a simple observation about Gaussian vectors.

² To avoid measurability issues, we do not formally define the distribution of a random process here. So the statement above should be understood as the fact that the covariance function determines the distribution of all marginals $(X_t)_{t \in T_0}$ with finite $T_0 \subset T$.

Lemma 7.1.12 (Gaussian random vectors). *Let Y be a mean zero Gaussian random vector in $\mathbb{R}^n$. Then there exist points $t_1, \dots, t_n \in \mathbb{R}^n$ such that*

$$Y \equiv (\langle g, t_i \rangle)_{i=1}^n, \quad \text{where } g \sim N(0, I_n).$$

Here “ $\equiv$ ” means that the distributions of the two random vectors are the same.

Proof Let Σ denote the covariance matrix of Y . Then we may realize

$$Y \equiv \Sigma^{1/2} g \quad \text{where } g \sim N(0, I_n)$$

(recall Section 3.3.2). Next, the coordinates of the vector $\Sigma^{1/2} g$ are $\langle t_i, g \rangle$ where t_i denote the rows of the matrix $\Sigma^{1/2}$. This completes the proof. $\square$

It follows that for any Gaussian process $(Y_s)_{s \in S}$, all finite-dimensional marginals $(Y_s)_{s \in S_0}$, $|S_0| = n$ can be represented as the canonical Gaussian process (7.1) indexed in a certain subset $T_0 \subset \mathbb{R}^n$.

Exercise 7.1.13. Realize an N -step random walk of Example 7.1.3 with $Z_i \sim N(0, 1)$ as a canonical Gaussian process (7.1) with $T \subset \mathbb{R}^N$. **Hint:** It might be simpler to think about increments $\|X_t - X_s\|_2$ instead of the covariance matrix.

7.2 Slepian's inequality

In many applications, it is useful to have a uniform control on a random process $(X_t)_{t \in T}$, i.e. to have a bound on³

$$\mathbb{E} \sup_{t \in T} X_t.$$

For some processes, this quantity can be computed exactly. For example, if (X_t) is a standard Brownian motion, the so-called *reflection principle* yields

$$\mathbb{E} \sup_{t \leq t_0} X_t = \sqrt{\frac{2t_0}{\pi}} \quad \text{for every } t_0 \geq 0.$$

For general random processes, even Gaussian, the problem is very non-trivial.

The first general bound we prove is Slepian's comparison inequality for Gaussian processes. It basically states that the faster the process grows (in terms of the magnitude of increments), the farther it gets.

Theorem 7.2.1 (Slepian's inequality). *Let $(X_t)_{t \in T}$ and $(Y_t)_{t \in T}$ be two mean zero Gaussian processes. Assume that for all $t, s \in T$, we have*

$$\mathbb{E} X_t^2 = \mathbb{E} Y_t^2 \quad \text{and} \quad \mathbb{E}(X_t - X_s)^2 \leq \mathbb{E}(Y_t - Y_s)^2. \quad (7.2)$$

Then for every $\tau \in \mathbb{R}$ we have

$$\mathbb{P} \left\{ \sup_{t \in T} X_t \geq \tau \right\} \leq \mathbb{P} \left\{ \sup_{t \in T} Y_t \geq \tau \right\}. \quad (7.3)$$

³ To avoid measurability issues, we study random processes through their finite-dimensional marginals as before. Thus we interpret $\mathbb{E} \sup_{t \in T} X_t$ more formally as $\sup_{T_0 \subset T} \mathbb{E} \max_{t \in T_0} X_t$ where the supremum is over all finite subsets $T_0 \subset T$.

Consequently,

$$\mathbb{E} \sup_{t \in T} X_t \leq \mathbb{E} \sup_{t \in T} Y_t. \quad (7.4)$$

Whenever the tail comparison inequality (7.3) holds, we say that the random variable X is *stochastically dominated* by the random variable Y .

We now prepare for the proof of Slepian's inequality.

7.2.1 Gaussian interpolation

The proof of Slepian's inequality that we are about to give will be based on the technique of *Gaussian interpolation*. Let us describe it briefly. Assume that T is finite; then $X = (X_t)_{t \in T}$ and $Y = (Y_t)_{t \in T}$ are Gaussian random vectors in $\mathbb{R}^n$ where $n = |T|$. We may also assume that X and Y are independent. (Why?)

Define the Gaussian random vector $Z(u)$ in $\mathbb{R}^n$ that continuously interpolates between $Z(0) = Y$ and $Z(1) = X$:

$$Z(u) := \sqrt{u} X + \sqrt{1-u} Y, \quad u \in [0, 1].$$

Exercise 7.2.2. Check that the covariance matrix of $Z(u)$ interpolates linearly between the covariance matrices of Y and X :

$$\Sigma(Z(u)) = u \Sigma(X) + (1-u) \Sigma(Y).$$

For a given function $f : \mathbb{R}^n \rightarrow \mathbb{R}$, we study how the quantity $\mathbb{E} f(Z(u))$ changes as u increases from 0 to 1. Of specific interest to us is the function

$$f(x) = \mathbf{1}_{\{\max_i x_i < \tau\}}.$$

We will be able to show that in this case, $\mathbb{E} f(Z(u))$ *increases* in u . This would imply the conclusion of Slepian's inequality at once, since then

$$\mathbb{E} f(Z(1)) \geq \mathbb{E} f(Z(0)), \quad \text{thus} \quad \mathbb{P} \left\{ \max_i X_i < \tau \right\} \geq \mathbb{P} \left\{ \max_i Y_i < \tau \right\}$$

as claimed.

Now let us pass to a detailed argument. To develop Gaussian interpolation, let us start with the following useful identity. In the results that follow, we implicitly assume that all expectations exist and are finite.

Lemma 7.2.3 (Gaussian integration by parts). *Let $X \sim N(0, 1)$. Then for any differentiable function $f : \mathbb{R} \rightarrow \mathbb{R}$ we have*

$$\mathbb{E} f'(X) = \mathbb{E} X f(X),$$

assuming both expectations exist and are finite.

Proof Assume first that f has bounded support. Denoting the Gaussian density of X by

$$p(x) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2},$$

we can express the expectation as an integral, and integrate it by parts:

$$\mathbb{E} f'(X) = \int_{\mathbb{R}} f'(x)p(x) dx = - \int_{\mathbb{R}} f(x)p'(x) dx. \quad (7.5)$$

Now, a direct check gives

$$p'(x) = -xp(x),$$

so the integral in (7.5) equals

$$\int_{\mathbb{R}} f(x)p(x)x dx = \mathbb{E} Xf(X),$$

as claimed. The identity can be extended to general functions by an approximation argument. The lemma is proved. $\square$

Exercise 7.2.4. 🍷 If $X \sim N(0, \sigma^2)$, show that

$$\mathbb{E} Xf(X) = \sigma^2 \mathbb{E} f'(X).$$

Hint: Represent $X = \sigma Z$ for $Z \sim N(0, 1)$, and apply Gaussian integration by parts.

Gaussian integration by parts generalizes nicely to high dimensions.

Lemma 7.2.5 (Multivariate Gaussian integration by parts). *Let $X \sim N(0, \Sigma)$. Then for any differentiable function $f : \mathbb{R}^n \rightarrow \mathbb{R}$ we have*

$$\mathbb{E} Xf(X) = \Sigma \cdot \mathbb{E} \nabla f(X),$$

assuming both expectations exist and are finite.

Exercise 7.2.6. 🍷🍷🍷 Prove Lemma 7.2.5. According to the matrix-by-vector multiplication, note that the conclusion of the lemma is equivalent to

$$\mathbb{E} X_i f(X) = \sum_{j=1}^n \Sigma_{ij} \mathbb{E} \frac{\partial f}{\partial x_j}(X), \quad i = 1, \dots, n. \quad (7.6)$$

Hint: Represent $X = \Sigma^{1/2}Z$ for $Z \sim N(0, I_n)$. Then

$$X_i = \sum_{k=1}^n (\Sigma^{1/2})_{ik} Z_k \quad \text{and} \quad \mathbb{E} X_i f(X) = \sum_{k=1}^n (\Sigma^{1/2})_{ik} \mathbb{E} Z_k f(\Sigma^{1/2}Z).$$

Apply univariate Gaussian integration by parts (Lemma 7.2.3) for $\mathbb{E} Z_k f(\Sigma^{1/2}Z)$ conditionally on all random variables except $Z_k \sim N(0, 1)$, and simplify.

Lemma 7.2.7 (Gaussian interpolation). *Consider two independent Gaussian random vectors $X \sim N(0, \Sigma^X)$ and $Y \sim N(0, \Sigma^Y)$. Define the interpolation Gaussian vector*

$$Z(u) := \sqrt{u} X + \sqrt{1-u} Y, \quad u \in [0, 1]. \quad (7.7)$$

Then for any twice-differentiable function $f : \mathbb{R}^n \rightarrow \mathbb{R}$, we have

$$\frac{d}{du} \mathbb{E} f(Z(u)) = \frac{1}{2} \sum_{i,j=1}^n (\Sigma_{ij}^X - \Sigma_{ij}^Y) \mathbb{E} \left[\frac{\partial^2 f}{\partial x_i \partial x_j}(Z(u)) \right], \quad (7.8)$$

assuming all expectations exist and are finite.

Proof Using the chain rule,⁴ we have

$$\begin{aligned} \frac{d}{du} \mathbb{E} f(Z(u)) &= \sum_{i=1}^n \mathbb{E} \frac{\partial f}{\partial x_i}(Z(u)) \frac{dZ_i}{du} \\ &= \frac{1}{2} \sum_{i=1}^n \mathbb{E} \frac{\partial f}{\partial x_i}(Z(u)) \left(\frac{X_i}{\sqrt{u}} - \frac{Y_i}{\sqrt{1-u}} \right) \quad (\text{by (7.7)}). \end{aligned} \quad (7.9)$$

Let us break this sum into two, and first compute the contribution of the terms containing X_i . To this end, we condition on Y and express

$$\sum_{i=1}^n \frac{1}{\sqrt{u}} \mathbb{E} X_i \frac{\partial f}{\partial x_i}(Z(u)) = \sum_{i=1}^n \frac{1}{\sqrt{u}} \mathbb{E} X_i g_i(X), \quad (7.10)$$

where

$$g_i(X) = \frac{\partial f}{\partial x_i}(\sqrt{u} X + \sqrt{1-u} Y).$$

Apply the multivariate Gaussian integration by parts (Lemma 7.2.5). According to (7.6), we have

$$\begin{aligned} \mathbb{E} X_i g_i(X) &= \sum_{j=1}^n \Sigma_{ij}^X \mathbb{E} \frac{\partial g_i}{\partial x_j}(X) \\ &= \sum_{j=1}^n \Sigma_{ij}^X \mathbb{E} \frac{\partial^2 f}{\partial x_i \partial x_j}(\sqrt{u} X + \sqrt{1-u} Y) \cdot \sqrt{u}. \end{aligned}$$

Substitute this into (7.10) to get

$$\sum_{i=1}^n \frac{1}{\sqrt{u}} \mathbb{E} X_i \frac{\partial f}{\partial x_i}(Z(u)) = \sum_{i,j=1}^n \Sigma_{ij}^X \mathbb{E} \frac{\partial^2 f}{\partial x_i \partial x_j}(Z(u)).$$

Taking expectation of both sides with respect to Y , we lift the conditioning on Y .

We can similarly evaluate the other sum in (7.9), the one containing the terms Y_i . Combining the two sums we complete the proof. $\square$

7.2.2 Proof of Slepian's inequality

We are ready to establish a preliminary, functional form Slepian's inequality.

Lemma 7.2.8 (Slepian's inequality, functional form). *Consider two mean zero Gaussian random vectors X and Y in $\mathbb{R}^n$. Assume that for all $i, j = 1, \dots, n$, we have*

$$\mathbb{E} X_i^2 = \mathbb{E} Y_i^2 \quad \text{and} \quad \mathbb{E}(X_i - X_j)^2 \leq \mathbb{E}(Y_i - Y_j)^2.$$

⁴ Here we use the multivariate chain rule to differentiate a function $f(g_1(u), \dots, g_n(u))$ where $g_i : \mathbb{R} \rightarrow \mathbb{R}$ and $f : \mathbb{R}^n \rightarrow \mathbb{R}$ as follows: $\frac{df}{du} = \sum_{i=1}^n \frac{\partial f}{\partial x_i} \frac{dg_i}{du}$.

Consider a twice-differentiable function $f : \mathbb{R}^n \rightarrow \mathbb{R}$ such that

$$\frac{\partial^2 f}{\partial x_i \partial x_j} \geq 0 \quad \text{for all } i \neq j.$$

Then

$$\mathbb{E} f(X) \geq \mathbb{E} f(Y),$$

assuming both expectations exist and are finite.

Proof The assumptions imply that the entries of the covariance matrices Σ^X and Σ^Y of X and Y satisfy

$$\Sigma_{ii}^X = \Sigma_{ii}^Y \quad \text{and} \quad \Sigma_{ij}^X \geq \Sigma_{ij}^Y.$$

for all $i, j = 1, \dots, n$. We can assume that X and Y are independent. (Why?) Apply Lemma 7.2.7 and using our assumptions, we conclude that

$$\frac{d}{du} \mathbb{E} f(Z(u)) \geq 0,$$

so $\mathbb{E} f(Z(u))$ increases in u . Then $\mathbb{E} f(Z(1)) = \mathbb{E} f(X)$ is at least as large as $\mathbb{E} f(Z(0)) = \mathbb{E} f(Y)$. This completes the proof. $\square$

Now we are ready to prove Slepian's inequality, Theorem 7.2.1. Let us state and prove it in the equivalent form for Gaussian random vectors.

Theorem 7.2.9 (Slepian's inequality). *Let X and Y be Gaussian random vectors as in Lemma 7.2.8. Then for every $\tau \geq 0$ we have*

$$\mathbb{P} \left\{ \max_{i \leq n} X_i \geq \tau \right\} \leq \mathbb{P} \left\{ \max_{i \leq n} Y_i \geq \tau \right\}.$$

Consequently,

$$\mathbb{E} \max_{i \leq n} X_i \leq \mathbb{E} \max_{i \leq n} Y_i.$$

Proof Let $h : \mathbb{R} \rightarrow [0, 1]$ be a twice-differentiable, non-increasing approximation to the indicator function of the interval $(-\infty, \tau)$:

$$h(x) \approx \mathbf{1}_{(-\infty, \tau)},$$

see Figure 7.2. Define the function $f : \mathbb{R}^n \rightarrow \mathbb{R}$ by

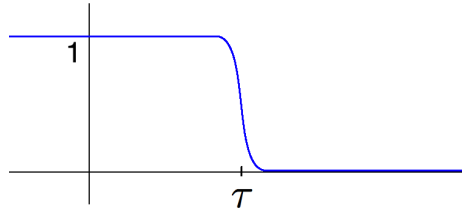


Figure 7.2 The function $h(x)$ is a smooth, non-increasing approximation to the indicator function $\mathbf{1}_{(-\infty, \tau)}$.

$$f(x) = h(x_1) \cdots h(x_n).$$

Then $f(x)$ is an approximation to the indicator function

$$f(x) \approx \mathbf{1}_{\{\max_i x_i < \tau\}}.$$

We are looking to apply the functional form of Slepian's inequality, Lemma 7.2.8, for $f(x)$. To check the assumptions of this result, note that for $i \neq j$ we have

$$\frac{\partial^2 f}{\partial x_i \partial x_j} = h'(x_i)h'(x_j) \cdot \prod_{k \notin \{i,j\}} h(x_k).$$

The first two factors are non-positive and the others are non-negative by the assumption. Thus the second derivative is non-negative, as required.

It follows that

$$\mathbb{E} f(X) \geq \mathbb{E} f(Y).$$

By approximation, this implies

$$\mathbb{P} \left\{ \max_{i \leq n} X_i < \tau \right\} \geq \mathbb{P} \left\{ \max_{i \leq n} Y_i < \tau \right\}.$$

This proves the first part of the conclusion. The second part follows using the integral identity in Lemma 1.2.1, see Exercise 7.2.10. $\square$

Exercise 7.2.10. Using the integral identity in Exercise 1.2.2, deduce the second part of Slepian's inequality (comparison of expectations).

7.2.3 Sudakov-Fernique's and Gordon's inequalities

Slepian's inequality has two assumptions on the processes (X_t) and (Y_t) in (7.2): the equality of variances and the dominance of increments. We now remove the assumption on the equality of variances, and still be able to obtain (7.4). This more practically useful result is due to Sudakov and Fernique.

Theorem 7.2.11 (Sudakov-Fernique's inequality). *Let $(X_t)_{t \in T}$ and $(Y_t)_{t \in T}$ be two mean zero Gaussian processes. Assume that for all $t, s \in T$, we have*

$$\mathbb{E}(X_t - X_s)^2 \leq \mathbb{E}(Y_t - Y_s)^2.$$

Then

$$\mathbb{E} \sup_{t \in T} X_t \leq \mathbb{E} \sup_{t \in T} Y_t.$$

Proof It is enough to prove this theorem for Gaussian random vectors X and Y in $\mathbb{R}^n$, just like we did for Slepian's inequality in Theorem 7.2.9. We again deduce the result from Gaussian Interpolation Lemma 7.2.7. But this time, instead of choosing $f(x)$ that approximates the indicator function of $\{\max_i x_i < \tau\}$, we want $f(x)$ to approximate $\max_i x_i$.

To this end, let $\beta > 0$ be a parameter and define the function⁵

$$f(x) := \frac{1}{\beta} \log \sum_{i=1}^n e^{\beta x_i}. \quad (7.11)$$

A quick check shows that

$$f(x) \rightarrow \max_{i \leq n} x_i \quad \text{as } \beta \rightarrow \infty.$$

(Do this!) Substituting $f(x)$ into the Gaussian interpolation formula (7.8) and simplifying the expression shows that $\frac{d}{du} \mathbb{E} f(Z(u)) \leq 0$ for all u (see Exercise 7.2.12 below). The proof can then be completed just like the proof of Slepian's inequality. $\square$

Exercise 7.2.12.  Show that $\frac{d}{du} \mathbb{E} f(Z(u)) \leq 0$ in Sudakov-Fernique's Theorem 7.2.11.

Hint: Differentiate f and check that

$$\frac{\partial f}{\partial x_i} = \frac{e^{\beta x_i}}{\sum_k e^{\beta x_k}} =: p_i(x) \quad \text{and} \quad \frac{\partial^2 f}{\partial x_i \partial x_j} = \beta (\delta_{ij} p_i(x) - p_i(x) p_j(x))$$

where δ_{ij} is the Kronecker delta, which equals 1 if $i = j$ and 0 otherwise. Next, check the following numeric identity:

$$\text{If } \sum_{i=1}^n p_i = 1 \quad \text{then} \quad \sum_{i,j=1}^n \sigma_{ij} (\delta_{ij} p_i - p_i p_j) = \frac{1}{2} \sum_{i \neq j} (\sigma_{ii} + \sigma_{jj} - 2\sigma_{ij}) p_i p_j.$$

Use Gaussian interpolation formula 7.2.7. Simplify the expression using the identity above with $\sigma_{ij} = \Sigma_{ij}^X - \Sigma_{ij}^Y$ and $p_i = p_i(Z(u))$. Deduce that

$$\frac{d}{du} \mathbb{E} f(Z(u)) = \frac{\beta}{4} \sum_{i \neq j} \left[\mathbb{E}(X_i - X_j)^2 - \mathbb{E}(Y_i - Y_j)^2 \right] \mathbb{E} p_i(Z(u)) p_j(Z(u)).$$

By the assumptions, this expression is non-positive.

Exercise 7.2.13 (Gaussian contraction inequality).  The following is a Gaussian version of Talagrand's contraction principle we proved in Exercise 6.7.7. Consider a bounded subset $T \subset \mathbb{R}^n$, and let $g_1, \dots, g_n$ be independent $N(0, 1)$ random variables. Let $\phi_i : \mathbb{R} \rightarrow \mathbb{R}$ be contractions, i.e. Lipschitz functions with $\|\phi_i\|_{\text{Lip}} \leq 1$. Prove that

$$\mathbb{E} \sup_{t \in T} \sum_{i=1}^n g_i \phi_i(t_i) \leq \mathbb{E} \sup_{t \in T} \sum_{i=1}^n g_i t_i.$$

Hint: Use Sudakov-Fernique's inequality.

Exercise 7.2.14 (Gordon's inequality).  Prove the following extension of Slepian's inequality due to Y. Gordon. Let $(X_{ut})_{u \in U, t \in T}$ and $Y = (Y_{ut})_{u \in U, t \in T}$ be

⁵ The motivation for considering this form of $f(x)$ comes from statistical mechanics, where the right side of (7.11) can be interpreted as a *log-partition function* and β as the *inverse temperature*.

two mean zero Gaussian processes indexed by pairs of points (u, t) in a product set $U \times T$. Assume that we have

$$\begin{aligned} \mathbb{E} X_{ut}^2 &= \mathbb{E} Y_{ut}^2, & \mathbb{E}(X_{ut} - X_{us})^2 &\leq \mathbb{E}(Y_{ut} - Y_{us})^2 & \text{for all } u, t, s; \\ \mathbb{E}(X_{ut} - X_{vs})^2 &\geq \mathbb{E}(Y_{ut} - Y_{vs})^2 & \text{for all } u \neq v \text{ and all } t, s. \end{aligned}$$

Then for every $\tau \geq 0$ we have

$$\mathbb{P} \left\{ \inf_{u \in U} \sup_{t \in T} X_{ut} \geq \tau \right\} \leq \mathbb{P} \left\{ \inf_{u \in U} \sup_{t \in T} Y_{ut} \geq \tau \right\}.$$

Consequently,

$$\mathbb{E} \inf_{u \in U} \sup_{t \in T} X_{ut} \leq \mathbb{E} \inf_{u \in U} \sup_{t \in T} Y_{ut}. \quad (7.12)$$

Hint: Use Gaussian Interpolation Lemma 7.2.7 for $f(x) = \prod_i [1 - \prod_j h(x_{ij})]$ where $h(x)$ is an approximation to the indicator function $\mathbf{1}_{\{x \leq \tau\}}$, as in the proof of Slepian's inequality.

Similarly to Sudakov-Fernique's inequality, it is possible to remove the assumption of equal variances from Gordon's theorem, and still be able to derive (7.12). We do not prove this result.

7.3 Sharp bounds on Gaussian matrices

We illustrate Gaussian comparison inequalities that we just proved with an application to random matrices. In Section 4.6, we studied $m \times n$ random matrices A with independent, sub-gaussian rows. We used the ε -net argument to control the norm of A as follows:

$$\mathbb{E} \|A\| \leq \sqrt{m} + C\sqrt{n}$$

where C is a constant. (See Exercise 4.6.3.) We now use Sudakov-Fernique's inequality to improve upon this bound for *Gaussian* random matrices, showing that it holds with sharp constant $C = 1$.

Theorem 7.3.1 (Norms of Gaussian random matrices). *Let A be an $m \times n$ matrix with independent $N(0, 1)$ entries. Then*

$$\mathbb{E} \|A\| \leq \sqrt{m} + \sqrt{n}.$$

Proof We can realize the norm of A as a supremum of a Gaussian process. Indeed,

$$\|A\| = \max_{u \in S^{n-1}, v \in S^{m-1}} \langle Au, v \rangle = \max_{(u,v) \in T} X_{uv}$$

where T denotes the product set $S^{n-1} \times S^{m-1}$ and

$$X_{uv} := \langle Au, v \rangle \sim N(0, 1).$$

(Check!)

To apply Sudakov-Fernique's comparison inequality (Theorem 7.2.11), let us compute the increments of the process (X_{uv}) . For any $(u, v), (w, z) \in T$, we have

$$\begin{aligned}\mathbb{E}(X_{uv} - X_{wz})^2 &= \mathbb{E}(\langle Au, v \rangle - \langle Aw, z \rangle)^2 = \mathbb{E}\left(\sum_{i,j} A_{ij}(u_j v_i - w_j z_i)\right)^2 \\ &= \sum_{i,j} (u_j v_i - w_j z_i)^2 \quad (\text{by independence, mean 0, variance 1}) \\ &= \|uv^\top - wz^\top\|_F^2 \\ &\leq \|u - w\|_2^2 + \|v - z\|_2^2 \quad (\text{see Exercise 7.3.2 below}).\end{aligned}$$

Let us define a simpler Gaussian process (Y_{uv}) with similar increments as follows:

$$Y_{uv} := \langle g, u \rangle + \langle h, v \rangle, \quad (u, v) \in T,$$

where

$$g \sim N(0, I_n), \quad h \sim N(0, I_m)$$

are independent Gaussian vectors. The increments of this process are

$$\begin{aligned}\mathbb{E}(Y_{uv} - Y_{wz})^2 &= \mathbb{E}(\langle g, u - w \rangle + \langle h, v - z \rangle)^2 \\ &= \mathbb{E}\langle g, u - w \rangle^2 + \mathbb{E}\langle h, v - z \rangle^2 \quad (\text{by independence, mean 0}) \\ &= \|u - w\|_2^2 + \|v - z\|_2^2 \quad (\text{since } g, h \text{ are standard normal}).\end{aligned}$$

Comparing the increments of the two processes, we see that

$$\mathbb{E}(X_{uv} - X_{wz})^2 \leq \mathbb{E}(Y_{uv} - Y_{wz})^2 \quad \text{for all } (u, v), (w, z) \in T,$$

as required in Sudakov-Fernique's inequality. Applying Theorem 7.2.11, we obtain

$$\begin{aligned}\mathbb{E}\|A\| &= \mathbb{E} \sup_{(u,v) \in T} X_{uv} \leq \mathbb{E} \sup_{(u,v) \in T} Y_{uv} \\ &= \mathbb{E} \sup_{u \in S^{n-1}} \langle g, u \rangle + \mathbb{E} \sup_{v \in S^{m-1}} \langle h, v \rangle \\ &= \mathbb{E}\|g\|_2 + \mathbb{E}\|h\|_2 \\ &\leq (\mathbb{E}\|g\|_2^2)^{1/2} + (\mathbb{E}\|h\|_2^2)^{1/2} \quad (\text{by inequality (1.3) for } L^p \text{ norms}) \\ &= \sqrt{n} + \sqrt{m} \quad (\text{recall Lemma 3.2.4}).\end{aligned}$$

This completes the proof. $\square$

Exercise 7.3.2.  Prove the following bound used in the proof of Theorem 7.3.1. For any vectors $u, w \in S^{n-1}$ and $v, z \in S^{m-1}$, we have

$$\|uv^\top - wz^\top\|_F^2 \leq \|u - w\|_2^2 + \|v - z\|_2^2.$$

While Theorem 7.3.1 does not give any tail bound for $\|A\|$, we can automatically deduce a tail bound using concentration inequalities we studied in Section 5.2.

Corollary 7.3.3 (Norms of Gaussian random matrices: tails). *Let A be an $m \times n$ matrix with independent $N(0, 1)$ entries. Then for every $t \geq 0$, we have*

$$\mathbb{P} \left\{ \|A\| \geq \sqrt{m} + \sqrt{n} + t \right\} \leq 2 \exp(-ct^2).$$

Proof This result follows by combining Theorem 7.3.1 with the concentration inequality in the Gauss space, Theorem 5.2.2.

To use concentration, let us view A as a long random vector in $\mathbb{R}^{m \times n}$ by concatenating the rows. This makes A a standard normal random vector, i.e. $A \sim N(0, I_{nm})$. Consider the function $f(A) := \|A\|$ that assigns to the vector A the operator norm of the matrix A . We have

$$f(A) \leq \|A\|_2,$$

where $\|A\|_2$ is the Euclidean norm in $\mathbb{R}^{m \times n}$. (This is the same as the Frobenius norm of A , which dominates the operator norm of A .) This shows that $A \mapsto \|A\|$ is a Lipschitz function on $\mathbb{R}^{m \times n}$, and its Lipschitz norm is bounded by 1. (Why?) Then Theorem 5.2.2 yields

$$\mathbb{P} \left\{ \|A\| \geq \mathbb{E} \|A\| + t \right\} \leq 2 \exp(-ct^2).$$

The bound on $\mathbb{E} \|A\|$ from Theorem 7.3.1 completes the proof. $\square$

Exercise 7.3.4 (Smallest singular values).  Use Gordon's inequality stated in Exercise 7.2.14 to obtain a sharp bound on the smallest singular value of an $m \times n$ random matrix A with independent $N(0, 1)$ entries:

$$\mathbb{E} s_n(A) \geq \sqrt{m} - \sqrt{n}.$$

Combine this result with concentration to show the tail bound

$$\mathbb{P} \left\{ \|A\| \leq \sqrt{m} - \sqrt{n} - t \right\} \leq 2 \exp(-ct^2).$$

Hint: Relate the smallest singular value to the min-max of a Gaussian process:

$$s_n(A) = \min_{u \in S^{n-1}} \max_{v \in S^{m-1}} \langle Au, v \rangle.$$

Apply Gordon's inequality (without the requirement of equal variances, which is noted below Exercise 7.2.14) to show that

$$\mathbb{E} s_n(A) \geq \mathbb{E} \|h\|_2 - \mathbb{E} \|g\|_2 \quad \text{where} \quad g \sim N(0, I_n), h \sim N(0, I_m).$$

Combine this with the fact that $f(n) := \mathbb{E} \|g\|_2 - \sqrt{n}$ is increasing in dimension n . (Take this fact for granted; it can be proved by a tedious calculation.)

Exercise 7.3.5 (Symmetric random matrices).  Modify the arguments above to bound the norm of a *symmetric* $n \times n$ Gaussian random matrix A whose entries above the diagonal are independent $N(0, 1)$ random variables, and the diagonal entries are independent $N(0, 2)$ random variables. This distribution of random matrices is called the *Gaussian orthogonal ensemble (GOE)*. Show that

$$\mathbb{E} \|A\| \leq 2\sqrt{n}.$$

Next, deduce the tail bound

$$\mathbb{P} \left\{ \|A\| \geq 2\sqrt{n} + t \right\} \leq 2 \exp(-ct^2).$$

7.4 Sudakov's minoration inequality

Let us return to studying general mean zero Gaussian processes $(X_t)_{t \in T}$. As we observed in Remark 7.1.7, the increments

$$d(t, s) := \|X_t - X_s\|_{L^2} = (\mathbb{E}(X_t - X_s)^2)^{1/2} \quad (7.13)$$

define a metric on the (otherwise abstract) index set T , which we called the *canonical metric*.

The canonical metric $d(t, s)$ determines the covariance function $\Sigma(t, s)$, which in turn determines the distribution of the process $(X_t)_{t \in T}$ (recall Exercise 7.1.8 and Remark 7.1.11.) So in principle, we should be able to answer any question about the distribution of a Gaussian process $(X_t)_{t \in T}$ by looking at the geometry of the metric space (T, d) . Put plainly, we should be able to study probability via geometry.

Let us then ask an important specific question. How can we evaluate the overall magnitude of the process, namely

$$\mathbb{E} \sup_{t \in T} X_t, \quad (7.14)$$

in terms of the geometry of (T, d) ? This turns out to be a difficult problem, which we start to study here and continue in Chapter 8.

In this section, we prove a useful lower bound on (7.14) in terms of the *metric entropy* of the metric space (T, d) . Recall from Section 4.2 that for $\varepsilon > 0$, the *covering number*

$$\mathcal{N}(T, d, \varepsilon)$$

is defined to be the smallest cardinality of an ε -net of T in the metric d . Equivalently, $\mathcal{N}(T, d, \varepsilon)$ is the smallest number⁶ of closed balls of radius ε whose union covers T . Recall also that the logarithm of the covering number,

$$\log_2 \mathcal{N}(T, d, \varepsilon)$$

is called the *metric entropy* of T .

Theorem 7.4.1 (Sudakov's minoration inequality). *Let $(X_t)_{t \in T}$ be a mean zero Gaussian process. Then, for any $\varepsilon \geq 0$, we have*

$$\mathbb{E} \sup_{t \in T} X_t \geq c\varepsilon \sqrt{\log \mathcal{N}(T, d, \varepsilon)}.$$

where d is the canonical metric defined in (7.13).

⁶ If T does not admit a finite ε -net, we set $\mathcal{N}(T, d, \varepsilon) = \infty$.

Proof Let us deduce this result from Sudakov-Fernique's comparison inequality (Theorem 7.2.11). Assume that

$$\mathcal{N}(T, d, \varepsilon) =: N$$

is finite; the infinite case will be considered in Exercise 7.4.2. Let $\mathcal{N}$ be a maximal ε -separated subset of T . Then $\mathcal{N}$ is an ε -net of T (recall Lemma 4.2.6), and thus

$$|\mathcal{N}| \geq N.$$

Restricting the process to $\mathcal{N}$, we see that it suffices to show that

$$\mathbb{E} \sup_{t \in \mathcal{N}} X_t \geq c\varepsilon \sqrt{\log N}.$$

We can do it by comparing $(X_t)_{t \in \mathcal{N}}$ to a simpler Gaussian process $(Y_t)_{t \in \mathcal{N}}$, which we define as follows:

$$Y_t := \frac{\varepsilon}{\sqrt{2}} g_t, \quad \text{where } g_t \text{ are independent } N(0, 1) \text{ random variables.}$$

To use Sudakov-Fernique's comparison inequality (Theorem 7.2.11), we need to compare the increments of the two processes. Fix two different points $t, s \in \mathcal{N}$. By definition, we have

$$\mathbb{E}(X_t - X_s)^2 = d(t, s)^2 \geq \varepsilon^2$$

while

$$\mathbb{E}(Y_t - Y_s)^2 = \frac{\varepsilon^2}{2} \mathbb{E}(g_t - g_s)^2 = \varepsilon^2.$$

(In the last line, we use that $g_t - g_s \sim N(0, 2)$.) This implies that

$$\mathbb{E}(X_t - X_s)^2 \geq \mathbb{E}(Y_t - Y_s)^2 \quad \text{for all } t, s \in \mathcal{N}.$$

Applying Theorem 7.2.11, we obtain

$$\mathbb{E} \sup_{t \in \mathcal{N}} X_t \geq \mathbb{E} \sup_{t \in \mathcal{N}} Y_t = \frac{\varepsilon}{\sqrt{2}} \mathbb{E} \max_{t \in \mathcal{N}} g_t \geq c\varepsilon \sqrt{\log N}.$$

In the last inequality we used that the expected maximum of N standard normal random variables is at least $c\sqrt{\log N}$, see Exercise 2.5.11. The proof is complete. $\square$

Exercise 7.4.2 (Sudakov's minoration for non-compact sets). ☛☛ Show that if (T, d) is not relatively compact, that is if $\mathcal{N}(T, d, \varepsilon) = \infty$ for some $\varepsilon > 0$, then

$$\mathbb{E} \sup_{t \in T} X_t = \infty.$$

7.4.1 Application for covering numbers in $\mathbb{R}^n$

Sudakov's minoration inequality can be used to estimate the covering numbers of sets $T \subset \mathbb{R}^n$. To see how to do this, consider a canonical Gaussian process on T , namely

$$X_t := \langle g, t \rangle, \quad t \in T, \quad \text{where } g \sim N(0, I_n).$$

As we observed in Section 7.1.2, the canonical distance for this process is the Euclidean distance in $\mathbb{R}^n$, i.e.

$$d(t, s) = \|X_t - X_s\|_{L^2} = \|t - s\|_2.$$

Thus Sudakov's inequality can be stated as follows.

Corollary 7.4.3 (Sudakov's minoration inequality in $\mathbb{R}^n$). *Let $T \subset \mathbb{R}^n$. Then, for any $\varepsilon > 0$, we have*

$$\mathbb{E} \sup_{t \in T} \langle g, t \rangle \geq c\varepsilon \sqrt{\log \mathcal{N}(T, \varepsilon)}.$$

Here $\mathcal{N}(T, \varepsilon)$ is the covering number of T by Euclidean balls – the smallest number of Euclidean balls with radii ε and centers in T that cover T , just like in Section 4.2.1.

To give an illustration of Sudakov's minoration, note that it yields (up to an absolute constant) the same bound on the covering numbers of polytopes in $\mathbb{R}^n$ that we gave in Corollary 0.0.4:

Corollary 7.4.4 (Covering numbers of polytopes). *Let P be a polytope in $\mathbb{R}^n$ with N vertices and whose diameter is bounded by 1. Then, for every $\varepsilon > 0$ we have*

$$\mathcal{N}(P, \varepsilon) \leq N^{C/\varepsilon^2}.$$

Proof As before, by translation, we may assume that the radius of P is bounded by 1. Denote by $x_1, \dots, x_N$ the vertices of P . Then

$$\mathbb{E} \sup_{t \in P} \langle g, t \rangle = \mathbb{E} \sup_{i \leq N} \langle g, x_i \rangle \leq C \sqrt{\log N}.$$

The equality here follows since the maximum of the linear function on the convex set P is attained at an extreme point, i.e. at a vertex of P . The bound is due to Exercise 2.5.10, since $\langle g, x \rangle \sim N(0, \|x\|_2^2)$ and $\|x\|_2 \leq 1$. Substituting this into Sudakov's minoration inequality of Corollary 7.4.3 and simplifying, we complete the proof. $\square$

Exercise 7.4.5 (Volume of polytopes).  Let P be a polytope in $\mathbb{R}^n$, which has N vertices and is contained in the unit Euclidean ball B_2^n . Show that

$$\frac{\text{Vol}(P)}{\text{Vol}(B_2^n)} \leq \left(\frac{C \log N}{n} \right)^{Cn}.$$

Hint: Use Proposition 4.2.12, Corollary 7.4.4 and optimize in ε .

7.5 Gaussian width

In the previous section, we encountered an important quantity associated with a general set $T \subset \mathbb{R}^n$. It is the magnitude of the canonical Gaussian process on T , i.e.

$$\mathbb{E} \sup_{t \in T} \langle g, t \rangle$$

where the expectation is taken with respect to the Gaussian random vector $g \sim N(0, I_n)$. This quantity plays a central role in high-dimensional probability and its applications. Let us give it a name and study its basic properties.

Definition 7.5.1. The *Gaussian width* of a subset $T \subset \mathbb{R}^n$ is defined as

$$w(T) := \mathbb{E} \sup_{x \in T} \langle g, x \rangle \quad \text{where } g \sim N(0, I_n).$$

One can think about Gaussian width $w(T)$ as one of the basic geometric quantities associated with subsets of $T \subset \mathbb{R}^n$, such as volume and surface area. Several variants of the definition of Gaussian width can be found in the literature, such as

$$\mathbb{E} \sup_{x \in T} |\langle g, x \rangle|, \quad \left(\mathbb{E} \sup_{x \in T} \langle g, x \rangle^2 \right)^{1/2}, \quad \mathbb{E} \sup_{x, y \in T} \langle g, x - y \rangle, \quad \text{etc.}$$

These versions are equivalent, or almost equivalent, to $w(T)$ as we will see in Section 7.6.

7.5.1 Basic properties

Proposition 7.5.2 (Gaussian width).

- (a) $w(T)$ is finite if and only if T is bounded.
- (b) Gaussian width is invariant under affine unitary transformations. Thus, for every orthogonal matrix U and any vector y , we have

$$w(UT + y) = w(T).$$

- (c) Gaussian width is invariant under taking convex hulls. Thus,

$$w(\text{conv}(T)) = w(T).$$

- (d) Gaussian width respects Minkowski addition of sets and scaling. Thus, for $T, S \subset \mathbb{R}^n$ and $a \in \mathbb{R}$ we have

$$w(T + S) = w(T) + w(S); \quad w(aT) = |a| w(T).$$

- (e) We have

$$w(T) = \frac{1}{2} w(T - T) = \frac{1}{2} \mathbb{E} \sup_{x, y \in T} \langle g, x - y \rangle.$$

(f) (Gaussian width and diameter). We have⁷

$$\frac{1}{\sqrt{2\pi}} \cdot \text{diam}(T) \leq w(T) \leq \frac{\sqrt{n}}{2} \cdot \text{diam}(T).$$

Proof Properties a–d are simple and will be checked in Exercise 7.5.3 below.

To prove property e, we use property d twice and get

$$w(T) = \frac{1}{2} [w(T) + w(T)] = \frac{1}{2} [w(T) + w(-T)] = \frac{1}{2} w(T - T),$$

as claimed.

To prove the lower bound in property f, fix a pair of points $x, y \in T$. Then both $x - y$ and $y - x$ lie in $T - T$, so by property e we have

$$\begin{aligned} w(T) &\geq \frac{1}{2} \mathbb{E} \max(\langle x - y, g \rangle, \langle y - x, g \rangle) \\ &= \frac{1}{2} \mathbb{E} |\langle x - y, g \rangle| = \frac{1}{2} \sqrt{\frac{2}{\pi}} \|x - y\|_2. \end{aligned}$$

The last identity follows since $\langle x - y, g \rangle \sim N(0, \|x - y\|_2^2)$ and since $\mathbb{E} |X| = \sqrt{2/\pi}$ for $X \sim N(0, 1)$. (Check!) It remains to take supremum over all $x, y \in T$, and the lower bound in property f follows.

To prove the upper bound in property f, we again use property e to get

$$\begin{aligned} w(T) &= \frac{1}{2} \mathbb{E} \sup_{x, y \in T} \langle g, x - y \rangle \\ &\leq \frac{1}{2} \mathbb{E} \sup_{x, y \in T} \|g\|_2 \|x - y\|_2 \leq \frac{1}{2} \mathbb{E} \|g\|_2 \cdot \text{diam}(T). \end{aligned}$$

It remains to recall that $\mathbb{E} \|g\|_2 \leq (\mathbb{E} \|g\|_2^2)^{1/2} = \sqrt{n}$. □

Exercise 7.5.3. ☹☹ Prove Properties a–d in Proposition 7.5.2.

Hint: Use rotation invariance of Gaussian distribution.

Exercise 7.5.4 (Gaussian width under linear transformations). ☹☹☹ Show that for any $m \times n$ matrix A , we have

$$w(AT) \leq \|A\| w(T).$$

Hint: Use Sudakov-Fernique's comparison inequality.

7.5.2 Geometric meaning of width

The notion of the Gaussian width of a set $T \subset \mathbb{R}^n$ has a nice geometric meaning. The width of T in the direction of a vector $\theta \in S^{n-1}$ is the smallest width of the slab that is formed by parallel hyperplanes orthogonal to θ and that contains T ; see Figure 7.3. Analytically, the width in the direction of θ can be expressed as

⁷ Recall that the diameter of a set $T \subset \mathbb{R}^n$ is defined as $\text{diam}(T) := \sup\{\|x - y\|_2 : x, y \in T\}$.

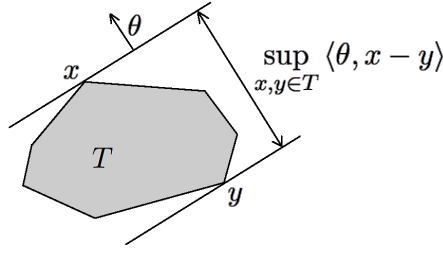


Figure 7.3 The width of a set $T \subset \mathbb{R}^n$ in the direction of a unit vector θ .

$$\sup_{x, y \in T} \langle \theta, x - y \rangle.$$

(Check!) If we average the width over all unit directions θ , we obtain the quantity

$$\mathbb{E} \sup_{x, y \in T} \langle \theta, x - y \rangle. \quad (7.15)$$

Definition 7.5.5 (Spherical width). The *spherical width*⁸ of a subset $T \subset \mathbb{R}^n$ is defined as

$$w_s(T) := \mathbb{E} \sup_{x \in T} \langle \theta, x \rangle \quad \text{where } \theta \sim \text{Unif}(S^{n-1}).$$

The quantity in (7.15) clearly equals $w_s(T - T)$.

How different are the Gaussian and spherical widths of T ? The difference is in the random vectors we use to do the averaging; they are $g \sim N(0, I_n)$ for Gaussian width and $\theta \sim \text{Unif}(S^{n-1})$ for spherical width. Both g and θ are rotation invariant, and, as we know, g is approximately $\sqrt{n}$ longer than θ . This makes Gaussian width just a scaling of the spherical width by approximately $\sqrt{n}$. Let us make this relation more precise.

Lemma 7.5.6 (Gaussian vs. spherical widths). *We have*

$$(\sqrt{n} - C) w_s(T) \leq w(T) \leq (\sqrt{n} + C) w_s(T).$$

Proof Let us express the Gaussian vector g through its length and direction:

$$g = \|g\|_2 \cdot \frac{g}{\|g\|_2} =: r\theta.$$

As we observed in Section 3.3.3, r and θ are independent and $\theta \sim \text{Unif}(S^{n-1})$. Thus

$$w(T) = \mathbb{E} \sup_{x \in T} \langle r\theta, x \rangle = (\mathbb{E} r) \cdot \mathbb{E} \sup_{x \in T} \langle \theta, x \rangle = \mathbb{E} \|g\|_2 \cdot w_s(T).$$

It remains to recall that concentration of the norm implies that

$$\left| \mathbb{E} \|g\|_2 - \sqrt{n} \right| \leq C,$$

see Exercise 3.1.4. □

⁸ The spherical width is also called the *mean width* in the literature.

7.5.3 Examples

Example 7.5.7 (Euclidean ball and sphere). The Gaussian width of the Euclidean unit sphere and ball is

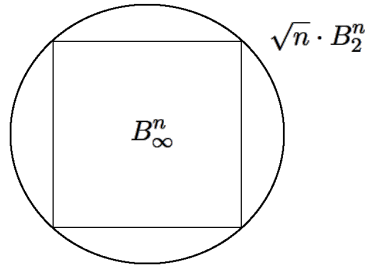
$$w(S^{n-1}) = w(B_2^n) = \mathbb{E} \|g\|_2 = \sqrt{n} \pm C, \quad (7.16)$$

where we used the result of Exercise 3.1.4. The spherical widths of these sets of course equal 1.

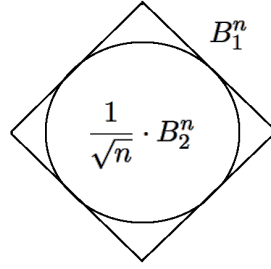
Example 7.5.8 (Cube). The unit ball of the ℓ_∞ norm in $\mathbb{R}^n$ is $B_\infty^n = [-1, 1]^n$. We have

$$\begin{aligned} w(B_\infty^n) &= \mathbb{E} \|g\|_1 \quad (\text{check!}) \\ &= \mathbb{E} |g_1| \cdot n = \sqrt{\frac{2}{\pi}} \cdot n. \end{aligned} \quad (7.17)$$

Comparing with (7.16), we see that Gaussian widths of the cube B_∞^n and its circumscribed ball $\sqrt{n}B_2^n$ have the same order n ; see Figure 7.4a.



(a) The Gaussian widths of the cube and its circumscribed ball are of the same order n .



(b) The Gaussian widths of B_1^n and its inscribed ball are almost of the same order.

Figure 7.4 Gaussian widths of some classical sets in $\mathbb{R}^n$.

Example 7.5.9 (ℓ_1 ball). The unit ball of the ℓ_1 norm in $\mathbb{R}^n$ is the set

$$B_1^n = \{x \in \mathbb{R}^n : \|x\|_1 \leq 1\}$$

which is sometimes called a *cross-polytope*; see Figure 7.5 for an illustration. The Gaussian width of the ℓ_1 ball can be bounded as follows:

$$c\sqrt{\log n} \leq w(B_1^n) \leq C\sqrt{\log n}. \quad (7.18)$$

To see this, check that

$$w(B_1^n) = \mathbb{E} \|g\|_\infty = \mathbb{E} \max_{i \leq n} |g_i|.$$

Then the bounds (7.18) follow from Exercises 2.5.10 and 2.5.11. Note that the Gaussian widths of the ℓ_1 ball B_1^n and its inscribed ball $\frac{1}{\sqrt{n}}B_2^n$ have almost same order (up to a logarithmic factor); see Figure 7.4b.

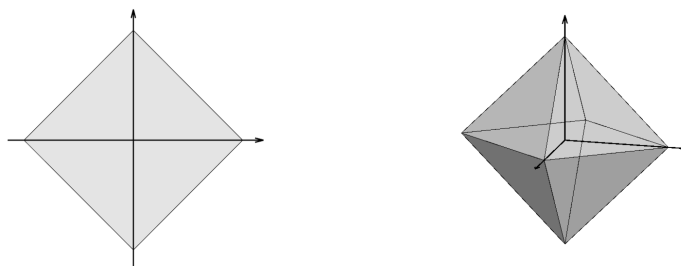


Figure 7.5 The unit ball of the ℓ_1 norm in $\mathbb{R}^n$, denoted B_1^n , is a diamond in dimension $n = 2$ (left) and a regular octahedron in dimension $n = 3$ (right).

Exercise 7.5.10 (Finite point sets). ☹ Let T be a finite set of points in $\mathbb{R}^n$. Check that

$$w(T) \leq C \sqrt{\log |T|} \cdot \text{diam}(T).$$

Hint: Argue like in the proof of Corollary 7.4.4.

Exercise 7.5.11 (ℓ_p balls). ☹☹☹ Let $1 \leq p < \infty$. Consider the unit ball of the ℓ_p norm in $\mathbb{R}^n$:

$$B_p^n := \{x \in \mathbb{R}^n : \|x\|_p \leq 1\}.$$

Check that

$$w(B_p^n) \leq C \sqrt{p'} n^{1/p'}.$$

Here p' denotes the *conjugate exponent* for p , which is defined by the equation $\frac{1}{p} + \frac{1}{p'} = 1$.

7.5.4 Surprising behavior of width in high dimensions

According to our computation in Example 7.5.9, the *spherical* width of B_1^n is

$$w_s(B_1^n) \asymp \sqrt{\frac{\log n}{n}}.$$

Surprisingly, it is much smaller than the diameter of B_1^n , which equals 2! Further, as we already noted, the Gaussian width of B_1^n is roughly the same (up to a logarithmic factor) as the Gaussian width of its inscribed Euclidean ball $\frac{1}{\sqrt{n}} B_2^n$. This again might look strange. Indeed, the cross-polytope B_1^n looks much larger than its inscribed ball whose diameter is $\frac{2}{\sqrt{n}}$! Why does Gaussian width behave this way?

Let us try to give an intuitive explanation. In high dimensions, the cube B_∞^n has so many vertices (2^n) that in most directions it extends to a radius close to $\sqrt{n}$, so it nearly covers the volume of the enclosing ball. In fact, the volumes of the cube and its circumscribed ball are both of the order C^n , so these sets are not far from each other from the volumetric point of view. So it should not be

very surprising to see that the Gaussian widths of the cube and its circumscribed ball are also of the same order.

The octahedron B_1^n has much fewer vertices ($2n$) than the cube. A random direction θ in $\mathbb{R}^n$ is likely to be almost orthogonal to all of them. So the width of B_1^n in the direction of θ is not significantly influenced by the presence of vertices. What really determines the width of B_1^n is its “bulk”, which is the inscribed Euclidean ball.

A similar picture can be seen from the volumetric viewpoint. There are so few vertices in B_1^n that the regions near them contain very little volume. The bulk of the volume of B_1^n lies much closer to the origin, not far from the inscribed Euclidean ball. Indeed, one can check that the volumes of B_1^n and its inscribed ball are both of the order of $(C/n)^n$. So from the volumetric point of view, the octahedron B_1^n is similar to its inscribed ball; Gaussian width gives the same result.

We can illustrate this phenomenon on Figure 7.6b that shows a “hyperbolic” picture of B_1^n that is due to V. Milman. Such pictures capture the bulk and outliers very well, but unfortunately they may not accurately show convexity.

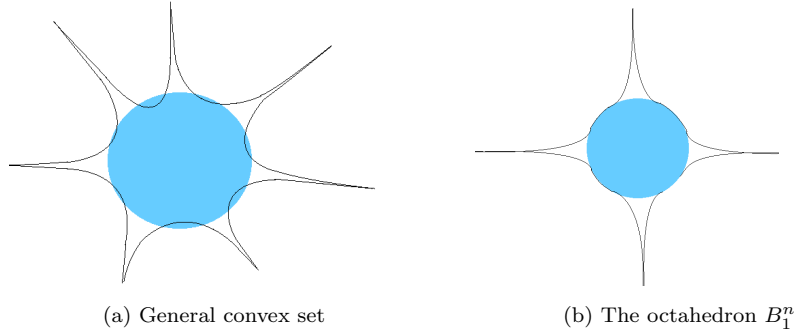


Figure 7.6 An intuitive, hyperbolic picture of a convex body in $\mathbb{R}^n$. The bulk is a round ball that contains most of the volume.

7.6 Stable dimension, stable rank, and Gaussian complexity

The notion of Gaussian width will help us to introduce a more robust version of the classical notion of dimension. The usual, linear algebraic, dimension $\dim T$ of a subset $T \subset \mathbb{R}^n$ is the smallest dimension of an affine subspace $E \subset \mathbb{R}^n$ that contains T . The linear algebraic dimension is unstable: it can significantly change (usually upwards) under a small perturbation of T . A more stable version of dimension can be defined based on the concept of Gaussian width.

In this section, it will be more convenient to work with a closely related *squared* version of the Gaussian width:

$$h(T)^2 := \mathbb{E} \sup_{t \in T} \langle g, t \rangle^2, \quad \text{where } g \sim N(0, I_n). \quad (7.19)$$

It is not difficult to see that the squared and usual versions of the Gaussian width are equivalent up to constant factor:

Exercise 7.6.1 (Equivalence). ☛☛☛ Check that

$$w(T - T) \leq h(T - T) \leq w(T - T) + C_1 \operatorname{diam}(T) \leq C w(T - T).$$

In particular, we have

$$2w(T) \leq h(T - T) \leq 2C w(T). \quad (7.20)$$

Hint: Use Gaussian concentration to prove the upper bound.

Definition 7.6.2 (Stable dimension). For a bounded set $T \subset \mathbb{R}^n$, the *stable dimension* of T is defined as

$$d(T) := \frac{h(T - T)^2}{\operatorname{diam}(T)^2} \asymp \frac{w(T)^2}{\operatorname{diam}(T)^2}.$$

The stable dimension is always bounded by the algebraic dimension:

Lemma 7.6.3. For any set $T \subset \mathbb{R}^n$, we have

$$d(T) \leq \dim(T).$$

Proof Let $\dim T = k$; this means that T lies in some subspace $E \subset \mathbb{R}^n$ of dimension k . By rotation invariance, we can assume that E is the coordinate subspace, i.e. $E = \mathbb{R}^k$. (Why?) By definition, we have

$$h(T - T)^2 = \mathbb{E} \sup_{x, y \in T} \langle g, x - y \rangle^2.$$

Since $x - y \in \mathbb{R}^k$ and $\|x - y\|_2 \leq \operatorname{diam}(T)$, we have $x - y = \operatorname{diam}(T) \cdot z$ for some $z \in B_2^k$. Thus the quantity above is bounded by

$$\operatorname{diam}(T)^2 \cdot \mathbb{E} \sup_{z \in B_2^k} \langle g, z \rangle^2 = \operatorname{diam}(T)^2 \cdot \mathbb{E} \|g'\|_2^2 = \operatorname{diam}(T)^2 \cdot k$$

where $g' \sim N(0, I_k)$ is a standard Gaussian random vector in $\mathbb{R}^k$. The proof is complete. $\square$

The inequality $d(T) \leq \dim(T)$ is in general sharp:

Exercise 7.6.4. ☛ Show that if T is a Euclidean ball in any subspace of $\mathbb{R}^n$, then

$$d(T) = \dim(T).$$

However, in many cases the stable dimension can be much smaller than the algebraic dimension:

Example 7.6.5. Let T be a finite set of points in $\mathbb{R}^n$. Then

$$d(T) \leq C \log |T|.$$

This follows from the bound on the Gaussian width of T in Exercise 7.5.10.

7.6.1 Stable rank

The stable dimension is more robust than the algebraic dimension. Indeed, small perturbation of a set T leads to small perturbation of Gaussian width and the diameter of T , and thus the stable dimension $d(T)$.

To give an example, consider the unit Euclidean ball B_2^n , for which both algebraic and stable dimensions equal n . Let us decrease one of the axes of B_2^n gradually from 1 to 0. The algebraic dimension will stay at n through this process and then instantly jump to $n - 1$. The stable dimension instead decreases gradually from n to $n - 1$. To see how exactly stable dimension decreases, do the following computation.

Exercise 7.6.6 (Ellipsoids). 🍷🍷 Let A be an $m \times n$ matrix, and let B_2^n denote the unit Euclidean ball. Check that the squared mean width of the ellipsoid AB_2^n is the Frobenius norm of A , i.e.

$$h(AB_2^n) = \|A\|_F.$$

Deduce that the stable dimension of the ellipsoid AB_2^n equals

$$d(AB_2^n) = \frac{\|A\|_F^2}{\|A\|^2}. \quad (7.21)$$

This example relates the stable dimension to the notion of *stable rank* of matrices, which is a robust version of the classical, linear algebraic rank.

Definition 7.6.7 (Stable rank). The *stable rank* of an $m \times n$ matrix A is defined as

$$r(A) := \frac{\|A\|_F^2}{\|A\|^2}.$$

The robustness of stable rank makes it a useful quantity in numerical linear algebra. The usual, algebraic, rank is the algebraic dimension of the image of A ; in particular

$$\text{rank}(A) = \dim(AB_2^n).$$

Similarly, (7.21) shows that the *stable rank* is the *statistical* dimension of the image:

$$r(A) = d(AB_2^n).$$

Finally, note that the stable rank is always bounded by the usual rank:

$$r(A) \leq \text{rank}(A).$$

(Check this!)

7.6.2 Gaussian complexity

Let us mention one more cousin of the Gaussian width where instead of squaring $\langle g, x \rangle$ as in (7.19) we take absolute value.

Definition 7.6.8. The *Gaussian complexity* of a subset $T \subset \mathbb{R}^n$ is defined as

$$\gamma(T) := \mathbb{E} \sup_{x \in T} |\langle g, x \rangle| \quad \text{where } g \sim N(0, I_n).$$

Obviously, we have

$$w(T) \leq \gamma(T),$$

and equality holds if T is origin-symmetric, i.e. if $T = -T$. Since $T - T$ is origin-symmetric, property e of Proposition 7.5.2 implies that

$$w(T) = \frac{1}{2}w(T - T) = \frac{1}{2}\gamma(T - T). \quad (7.22)$$

In general, Gaussian width and complexity may be quite different. For example, if T consists of a single nonzero point, then $w(T) = 0$ but $\gamma(T) > 0$. Still, these two quantities are very closely related:

Exercise 7.6.9 (Gaussian width vs. Gaussian complexity). ☛☛☛ Consider a set $T \subset \mathbb{R}^n$ and a point $y \in T$. Show that

$$\frac{1}{3} [w(T) + \|y\|_2] \leq \gamma(T) \leq 2 [w(T) + \|y\|_2]$$

This implies in particular that Gaussian width and Gaussian complexity are equivalent for any set T that contains the origin:

$$w(T) \leq \gamma(T) \leq 2w(T).$$

(It is fine if you prove the inequalities above with other absolute constants instead of 2 and 1/3.)

7.7 Random projections of sets

This section will illustrate the importance of the notion of Gaussian (and spherical) width in dimension reduction problems. Consider a set $T \subset \mathbb{R}^n$ and project it onto a random m -dimensional subspace in $\mathbb{R}^n$ (chosen uniformly from the Grassmannian $G_{n,m}$); see Figure 5.2 for illustration. In applications, we might think of T as a data set and P as a means of dimension reduction. What can we say about the size (diameter) of the projected set PT ?

For a *finite* set T , Johnson-Lindenstrauss Lemma (Theorem 5.3.1) states that as long as

$$m \gtrsim \log |T|, \quad (7.23)$$

the random projection P acts essentially as a scaling of T . Namely, P shrinks all distances between points in T by a factor $\approx \sqrt{m/n}$. In particular,

$$\text{diam}(PT) \approx \sqrt{\frac{m}{n}} \text{diam}(T). \quad (7.24)$$

If the cardinality of T is too large or infinite, then (7.24) may fail. For example,

if $T = B_2^n$ is a Euclidean ball then no projection can shrink the size of T at all, and we have

$$\text{diam}(PT) = \text{diam}(T). \quad (7.25)$$

What happens for a general set T ? The following result states that a random projection shrinks T as in (7.24), but it can not shrink it beyond the spherical width of T .

Theorem 7.7.1 (Sizes of random projections of sets). *Consider a bounded set $T \subset \mathbb{R}^n$. Let P be a projection in $\mathbb{R}^n$ onto a random m -dimensional subspace $E \sim \text{Unif}(G_{n,m})$. Then, with probability at least $1 - 2e^{-m}$, we have*

$$\text{diam}(PT) \leq C \left[w_s(T) + \sqrt{\frac{m}{n}} \text{diam}(T) \right].$$

To prove of this result, we pass to an equivalent probabilistic model, just like we did in the proof of Johnson-Lindenstrauss Lemma (see the proof of Proposition 5.3.2). First, a random subspace $E \subset \mathbb{R}^n$ can be realized by a random rotation of some fixed subspace, such as $\mathbb{R}^m$. Next, instead of fixing T and randomly rotating the subspace, we can fix the subspace and randomly rotate T . The following exercise makes this reasoning more formal.

Exercise 7.7.2 (Equivalent models for random projections). ☛☛ Let P be a projection in $\mathbb{R}^n$ onto a random m -dimensional subspace $E \sim \text{Unif}(G_{n,m})$. Let Q be an $m \times n$ matrix obtained by choosing the first m rows of a random $n \times n$ matrix $U \sim \text{Unif}(O(n))$ drawn uniformly from the orthogonal group.

- (a) Show that for any fixed point $x \in \mathbb{R}^n$,

$$\|Px\|_2 \text{ and } \|Qx\|_2 \text{ have the same distribution.}$$

Hint: Use the singular value decomposition of P .

- (b) Show that for any fixed point $z \in S^{m-1}$,

$$Q^T z \sim \text{Unif}(S^{n-1}).$$

In other words, the map Q^T acts as a random isometric embedding of $\mathbb{R}^m$ into $\mathbb{R}^n$. **Hint:** It is enough to check the rotation invariance of the distribution of $Q^T z$.

Proof of Theorem 7.7.1. Our argument is another example of an ε -net argument. Without loss of generality, we may assume that $\text{diam}(T) \leq 1$. (Why?)

Step 1: Approximation. By Exercise 7.7.2, it suffices to prove the theorem for Q instead of P . So we are going to bound

$$\text{diam}(QT) = \sup_{x \in T-T} \|Qx\|_2 = \sup_{x \in T-T} \max_{z \in S^{m-1}} \langle Qx, z \rangle.$$

Similarly to our older arguments (for example, in the proof of Theorem 4.4.5 on random matrices), we discretize the sphere S^{m-1} . Choose an $(1/2)$ -net $\mathcal{N}$ of S^{m-1} so that

$$|\mathcal{N}| \leq 5^m;$$

this is possible to do by Corollary 4.2.13. We can replace the supremum over the sphere S^{m-1} by the supremum over the net $\mathcal{N}$ paying a factor 2:

$$\text{diam}(QT) \leq 2 \sup_{x \in T-T} \max_{z \in \mathcal{N}} \langle Qx, z \rangle = 2 \max_{z \in \mathcal{N}} \sup_{x \in T-T} \langle Q^\top z, x \rangle. \quad (7.26)$$

(Recall Exercise 4.4.2.) We first control the quantity

$$\sup_{x \in T-T} \langle Q^\top z, x \rangle \quad (7.27)$$

for a fixed $z \in \mathcal{N}$ and with high probability, and then take union bound over all z .

Step 2: Concentration. So, let us fix $z \in \mathcal{N}$. By Exercise 7.7.2, $Q^\top z \sim \text{Unif}(S^{n-1})$. The expectation of (7.27) can be realized as the spherical width:

$$\mathbb{E} \sup_{x \in T-T} \langle Q^\top z, x \rangle = w_s(T - T) = 2w_s(T).$$

(The last identity is the spherical version of a similar property of the Gaussian width, see part e of Proposition 7.5.2.)

Next, let us check that (7.27) concentrates nicely around its mean $2w_s(T)$. For this, we can use the concentration inequality (5.6) for Lipschitz functions on the sphere. Since we assumed that $\text{diam}(T) \leq 1$ in the beginning, one can quickly check that the function

$$\theta \mapsto \sup_{x \in T-T} \langle \theta, x \rangle$$

is a Lipschitz function on the sphere S^{n-1} , and its Lipschitz norm is at most 1. (Do this!) Therefore, applying the concentration inequality (5.6), we obtain

$$\mathbb{P} \left\{ \sup_{x \in T-T} \langle Q^\top z, x \rangle \geq 2w_s(T) + t \right\} \leq 2 \exp(-cnt^2).$$

Step 3: Union bound. Now we unfix $z \in \mathcal{N}$ by taking the union bound over $\mathcal{N}$. We get

$$\mathbb{P} \left\{ \max_{z \in \mathcal{N}} \sup_{x \in T-T} \langle Q^\top z, x \rangle \geq 2w_s(T) + t \right\} \leq |\mathcal{N}| \cdot 2 \exp(-cnt^2) \quad (7.28)$$

Recall that $|\mathcal{N}| \leq 5^m$. Then, if we choose

$$t = C \sqrt{\frac{m}{n}}$$

with C large enough, the probability in (7.28) can be bounded by $2e^{-m}$. Then (7.28) and (7.26) yield

$$\mathbb{P} \left\{ \frac{1}{2} \text{diam}(QT) \geq 2w_s(T) + C \sqrt{\frac{m}{n}} \right\} \leq e^{-m}.$$

This proves Theorem 7.7.1. □

Exercise 7.7.3 (Gaussian projection). 🍄🍄🍄 Prove a version of Theorem 7.7.1 for $m \times n$ Gaussian random matrix G with independent $N(0, 1)$ entries. Specifically, show that for any bounded set $T \subset \mathbb{R}^n$, we have

$$\text{diam}(GT) \leq C \left[w(T) + \sqrt{m} \text{diam}(T) \right]$$

with probability at least $1 - 2e^{-m}$. Here $w(T)$ is the Gaussian width of T .

Exercise 7.7.4 (The reverse bound). 🍄🍄🍄 Show that the bound in Theorem 7.7.1 is optimal: prove the reverse bound

$$\mathbb{E} \text{diam}(PT) \geq c \left[w_s(T) + \sqrt{\frac{m}{n}} \text{diam}(T) \right]$$

for all bounded sets $T \subset \mathbb{R}^n$.

Hint: To obtain the bound $\mathbb{E} \text{diam}(PT) \gtrsim w_s(T)$, reduce P to a one-dimensional projection by dropping terms from the singular value decomposition of P . To obtain the bound $\mathbb{E} \text{diam}(PT) \geq \sqrt{\frac{m}{n}} \text{diam}(T)$, argue about a pair of points in T .

Exercise 7.7.5 (Random projections of matrices). 🍄🍄 Let A be an $n \times k$ matrix.

- (a) Let P be a projection in $\mathbb{R}^n$ onto a random m -dimensional subspace chosen uniformly in $G_{n,m}$. Show that with probability at least $1 - 2e^{-m}$, we have

$$\|PA\| \leq C \left[\frac{1}{\sqrt{n}} \|A\|_F + \sqrt{\frac{m}{n}} \|A\| \right].$$

- (b) Let G be an $m \times n$ Gaussian random matrix with independent $N(0, 1)$ entries. Show that with probability at least $1 - 2e^{-m}$, we have

$$\|GA\| \leq C \left(\|A\|_F + \sqrt{m} \|A\| \right).$$

Hint: Express the operator norm of PA to the diameter of the ellipsoid $P(AB_2^k)$, and use Theorem 7.7.1 in part 1 and Exercise 7.7.3 in part 2.

7.7.1 The phase transition

Let us pause to take a closer look at the bound Theorem 7.7.1 gives. We can equivalently write it as

$$\text{diam}(PT) \leq C \max \left[w_s(T), \sqrt{\frac{m}{n}} \text{diam}(T) \right].$$

Let us compute the dimension m for which the phase transition occurs between the two terms $w_s(T)$ and $\sqrt{\frac{m}{n}} \text{diam}(T)$. Setting them equal to each other and

solving for m , we find that the phase transition happens when

$$\begin{aligned} m &= \frac{(\sqrt{n} w_s(T))^2}{\text{diam}(T)^2} \\ &\asymp \frac{w(T)^2}{\text{diam}(T)^2} \quad (\text{pass to Gaussian width using Lemma 7.5.6}) \\ &\asymp d(T) \quad (\text{by Definition 7.6.2 of stable dimension}). \end{aligned}$$

So we can express the conclusion of Theorem 7.7.1 as follows:

$$\text{diam}(PT) \leq \begin{cases} C \sqrt{\frac{m}{n}} \text{diam}(T), & \text{if } m \geq d(T) \\ C w_s(T), & \text{if } m \leq d(T). \end{cases}$$

Figure 7.7 shows a graph of $\text{diam}(PT)$ as a function of the dimension m .

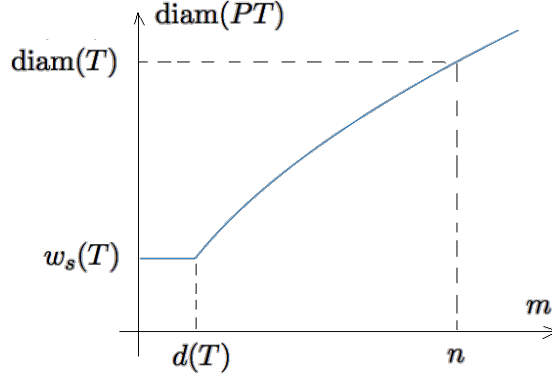


Figure 7.7 The diameter of a random m -dimensional projection of a set T as a function of m .

For large m , the random m -dimensional projection shrinks T by the factor $\sim \sqrt{m/n}$, just like we have seen in (7.24) in the context of Johnson-Lindenstrauss lemma. However, when the dimension m drops below the stable dimension $d(T)$, the shrinking stops – it levels off at the spherical width $w_s(T)$. We saw an example of this in (7.25), where a Euclidean ball can not be shrunk by a projection.

7.8 Notes

There are several introductory books on random processes (also called stochastic processes) and in particular on Brownian motion, for example [37, 127, 182, 156].

Slepian's inequality (Theorem 7.2.1) is originally due to D. Slepian [186, 187]; modern proofs can be found e.g. in [130, Corollary 3.12], [3, Section 2.2], [212, Section 6.1], [105], [111]. Sudakov-Fernique inequality (Theorem 7.2.11) is attributed to V. N. Sudakov [193, 194] and X. Fernique [75]. Our presentation of the proofs

of Slepian's and Sudakov-Fernique's inequalities in Section 7.2 is based on the approach of J.-P. Kahane [111] and a smoothing argument of S. Chatterjee (see [3, Section 2.2]), and it follows [212, Section 6.1]. A more general version of Gaussian contraction inequality in Exercise 7.2.13 can be found in [130, Corollary 3.17].

Gordon's inequality we stated in Exercise 7.2.14 and its extensions can be found in [84, 85, 88, 111]. Applications of Gordon's inequality for convex optimization can be found e.g. in [202, 200, 203].

The relevance of comparison inequalities to random matrix theory was noticed by S. Szarek. The applications we presented in Section 7.3 can be derived from the work of Y. Gordon [84]. Our presentation there follows the argument in [59, Section II.c], which is also reproduced in [222, Section 5.3.1].

Sudakov's minoration inequality (Theorem 7.4.1) was originally proved by V. N. Sudakov. Our presentation follows [130, Theorem 3.18]; see [11, Section 4.2] for an alternative proof via duality. The volume bound in Exercise 7.4.5 is almost best possible, but not quite. A slightly stronger bound

$$\frac{\text{Vol}(P)}{\text{Vol}(B_2^n)} \leq \left(\frac{C \log(1 + N/n)}{n} \right)^{n/2}$$

can be deduced in exactly the same way, if we use from the stronger bound on the covering numbers given in Exercise 0.0.6. This result is known and is best possible up to a constant C [50, Section 3].

Gaussian width and its cousins, which we introduce in Section 7.5, was originally introduced in geometric functional analysis and asymptotic convex geometry [11, 150]. More recently, starting from [178], the role of Gaussian width was recognized in applications to signal processing and high-dimensional statistics [189, 161, 190, 52, 169, 9, 201, 163]; see also [223, Section 3.5], [132]. In Section 7.5.4 we noted some surprising geometric phenomena in high dimensions; to learn more about them see the preface of [11] and [13].

The notion of stable dimension $d(T)$ of a set $T \subset \mathbb{R}^n$ introduced in Section 7.6 seems to be new. In the special case where T is a closed convex cone, the squared version of Gaussian width $h(T)$ defined in (7.19) is often called *statistical dimension* of T in the literature on signal recovery [140, 9, 163].

The notion of stable rank $r(A) = \|A\|_F^2 / \|A\|^2$ of a matrix A (also called effective, or numerical rank) seems to appear for the first time in [177]. In some literature (e.g. [222, 119]) the quantity

$$k(\Sigma) = \frac{\text{tr}(\Sigma)}{\|\Sigma\|}$$

is also called the stable rank of a symmetric positive semidefinite matrix Σ ; we call $k(\Sigma)$ the *intrinsic dimension* following [207, Definition 7.1.1]. Note that we used the quantity $k(\Sigma)$ in covariance estimation (see Remark 5.6.3). Clearly, if $\Sigma = A^T A$ or $\Sigma = A A^T$ then

$$k(\Sigma) = r(A).$$

Theorem [7.7.1](#) and its improvement that we will give in Section [9.2.2](#) is due to V. Milman [[149](#)], see also [[11](#), Proposition 5.7.1].